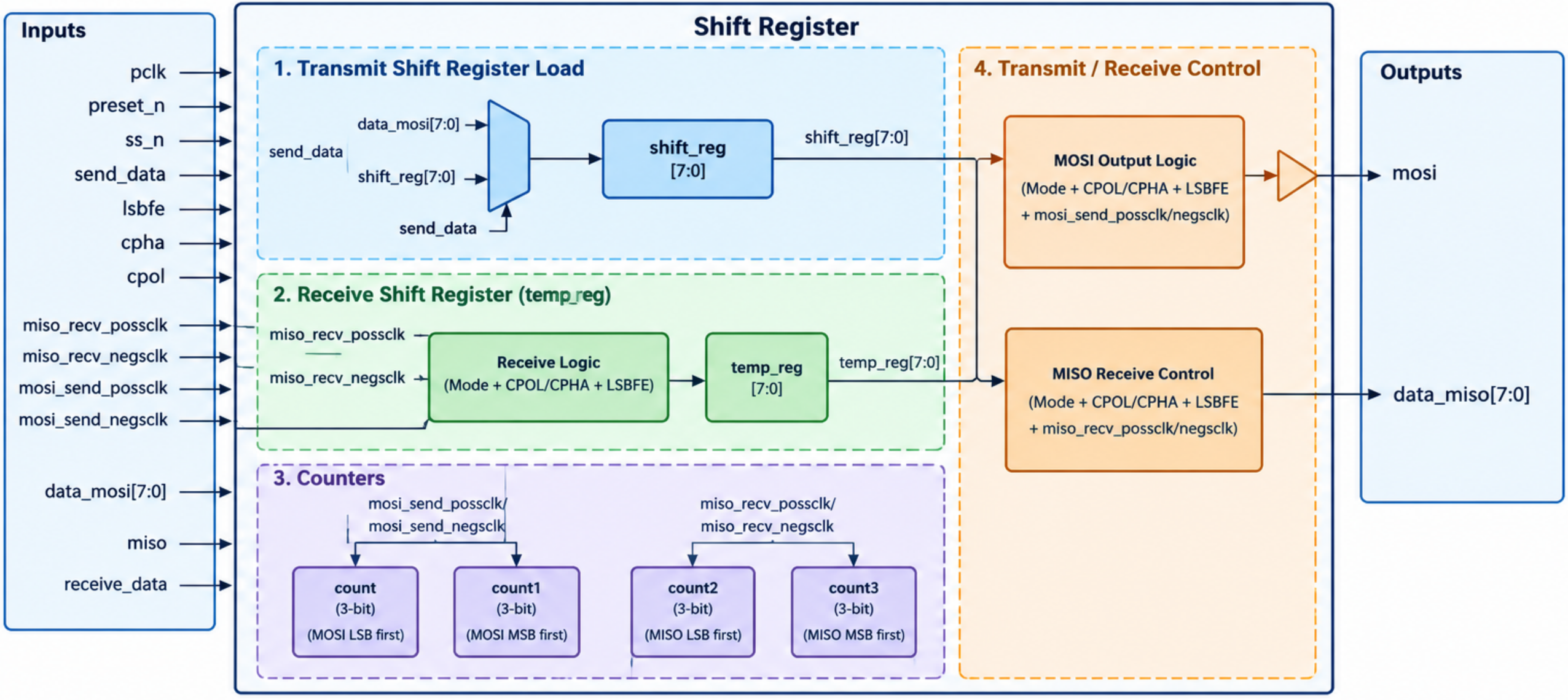
Shift Register – Block Diagram



Key Signals

- shift_reg – transmit data register
- temp_reg – receive data register
- count/count1 – MOSI bit counters
- count2/count3 – MISO bit counters
- data_miso – output receive data (when receive_date=1)

Shift Direction

- lsbfe = 1 → LSB first (count / count2)
- lsbfe = 0 → MSB first (count1 / count3)

SPI Mode Selection

- (!cpol && !cpha) → Mode 0
- (cpol && cpha) → Mode 0
- (!cpol && cpha) → Mode 1
- (cpol && !cpha) → Mode 1

(Used to select which edge signal to use for MOSI/MISO)

MOSI / MISO Edge Signals

- mosi_send_possclk – send on posedge
- mosi_send_negsclk – send on negedge
- miso_recv_possclk – receive on posedge
- miso_recv_negsclk – receive on negedge